Programming with Class VII and VIII

Coding Questions (LAQ)

1. ***Smart Employee Payroll System:*** *A company wants a payroll system, Requirements:*
   a. *Create an Employee class*
   b. *Use initializer lists*
   c. *Employee attributes: id, name, hourlyRate*
   d. *Throw an exception if hourlyRate is negative*
   e. *Display employee information*

2. ***University Student Registration System:*** *Requirements:*
   a. *Create Address class*
   b. *Create Student class*
   c. *Student contains Address*
   d. *Use composition and initializer lists*
   e. *Display student information*

Coding Questions (SAQ)

3. ***Initializer List:***
   a. *Create a Product class that stores product ID, product name. Use an initializer list.*
   b. *A patient has an address. Complete the constructor using composition.*

```
class Address
{
private:
    std::string city_;

public:
    Address(std::string city)
        : city_{std::move(city)}
    {
    }
};

class Patient
{
private:
    std::string patient_name_;
    Address address_;

public:
    Patient(
        std::string patient_name,
        std::string city);
};
```

4. ***Exception Handling:***
    a. *Create a custom exception class called InsufficientBalanceException*
    b. *Write a function that throws an exception if delivery distance is negative*
    c. *A patient has an address. Complete the constructor using composition.*

```
class Address
{
private:
    std::string city_;

public:
    Address(std::string city)
        : city_{std::move(city)}
    {
    }
};

class Patient
{
private:
    std::string patient_name_;
    Address address_;

public:
    Patient(
        std::string patient_name,
        std::string city);
};
```